

Erasmus Mundus Joint Master Artificial Intelligence for
Sustainable Societies (AISS)

Topics on Machine Learning and its Applications

Data representation

Pedro Strecht



Data representation

Terminology

Data types: numerical and categorical

Other types of data

Datasets

Terminology



Example (e_i)

- A single real-world entity represented as a tuple of values

$$e_i = (v_{i,1}, v_{i,2}, \dots, v_{i,d})$$

- Each example is one observation with multiple variable values



Variable (x_j)

- A fundamental unit of data capturing specific characteristics
- Represent quantifiable or observable properties of the entities under analysis
- For instance, in an individual, variables could include age, income level, gender, or educational attainment

d = total number of variables

$v_{\{i,j\}}$ = value of variable x_j for example e_i

Terminology



Example

- instance
- observation
- object
- tuple
- data point
- statistical unit



Variable

- attribute
- dimension
- feature

In this course, we will predominantly employ "example" and "variable" in both singular and plural forms.

Data Types



Quantitative (Numerical) Data

- Measurable quantities with numerical values
- Mathematical operations can be performed
- Can be continuous or discrete



Qualitative (Categorical) Data

- Descriptive characteristics or categories
- Represents qualities or attributes
- Can be nominal or ordinal



Numerical Data Types

Continuous Data

- Can assume any number within an interval (decimals, fractions)
- Result from measurements; can be refined indefinitely
- Represents phenomena varying smoothly in space
- **Examples:** temperature, elevation, weight, travel time, annual precipitation



Numerical Data Types

Discrete Data

- Assumes distinct, countable integers (counts)
- No intermediate values between consecutive numbers
- Refers to counts of entities
- **Examples:** Number of buildings, tree species count, person's age (in years), vehicle count at intersection



Categorical Data Types

Nominal Data

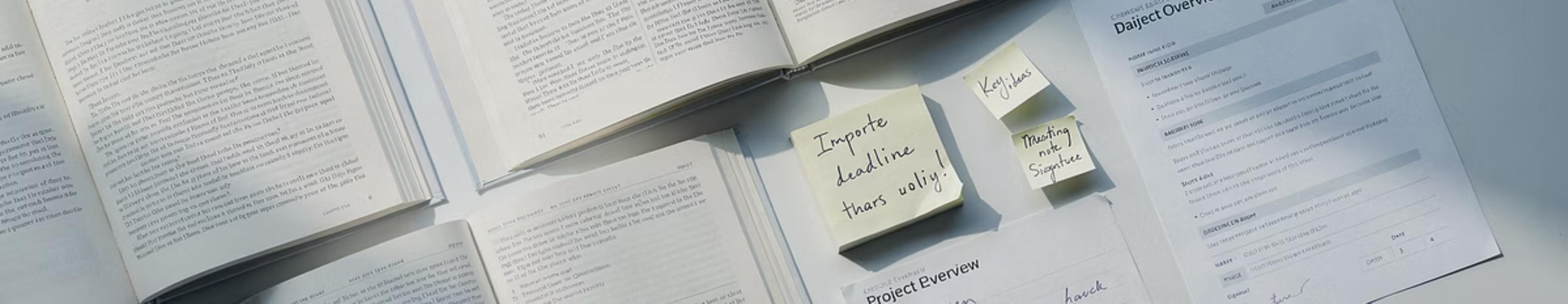
- Categories with no inherent order or hierarchy
- Difference is only in name, not magnitude
- **Examples:** Colors, gender, city names, types of land use (residential, commercial, industrial), tree species



Categorical Data Types

Ordinal Data

- Categories with a meaningful order or sequence
- Differences between values are not quantifiable or uniform
- **Examples:** Education level (elementary, secondary, higher), satisfaction rating (poor, fair, good), risk categories (low, medium, high), disaster severity degree



Other types of data

Textual data

- Unstructured data from human language, requiring natural language processing
- Includes written documents, social media posts, emails, and articles
- **Examples:** Book chapters, customer reviews, tweets, news articles, transcripts



Other types of data

Temporal data

- Data points associated with specific times, often used for trend analysis
- Captures changes and patterns over time periods
- **Examples:** Stock prices over time, weather measurements by date, website traffic logs, sensor readings with timestamps

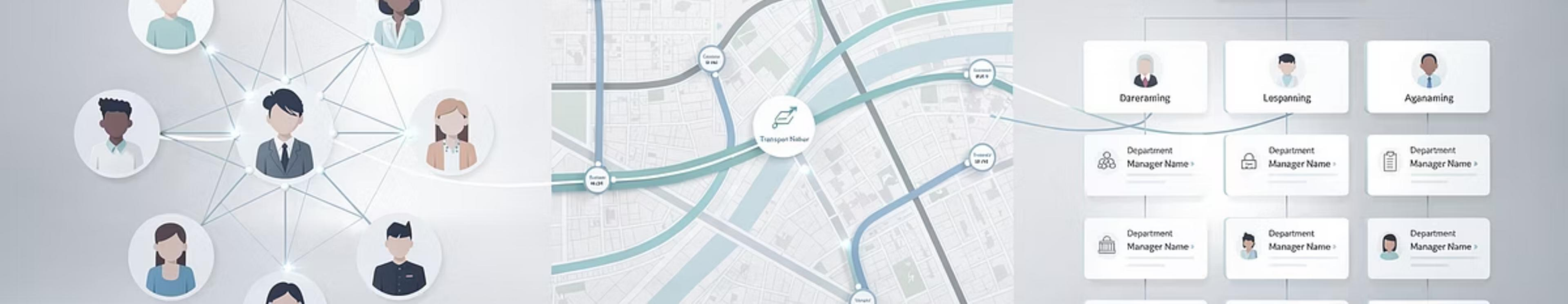


Other types of data

Spatial data

Information tied to geographic locations, visualized on maps

- **Geographic data** – Data associated with geographic locations, like GPS coordinates or maps
- **Geospatial data** – Data that represents spatial relationships between objects, such as the distance between cities or the overlap of regions



Other types of data

Network data

Data representing relationships and connections between entities in a network structure

- **Graph data** – Nodes and edges representing entities and their relationships, like social networks or transportation systems
- **Relational data** – Data showing how different entities interact or connect with each other



Other types of data

Image data

Digital images captured or created electronically, containing pixel-based visual information

- **Image data** – Digital images, such as photographs, scans, or digitally generated graphics
- Common formats include JPEG, PNG, TIFF, and RAW files, each with different compression and quality characteristics
- For ML applications, images are represented as multi-dimensional arrays (tensors) where each pixel contains numerical values for color channels (e.g., RGB values from 0-255)



Other types of data

Audio data

Sound recordings, representing speech, music, or environmental sounds.

- **Speech data** – Audio recordings of spoken language
- **Music data** – Audio recordings of music

Datasets

- Datasets are structured collections of information, organized for efficient analysis
- They can include diverse data types, but are typically arranged in a row and column format, where rows represent individual examples and columns represent variables or features

1

Rows (Examples)

Each row typically represents a single record or observation within the dataset

2

Columns (Variables)

Each column represents a specific attribute, feature, or measurement for all records

3

Structured for Analysis

This format allows for efficient querying, statistical analysis, and machine learning model training

Datasets

Rows (e_i)

Each row represents a single example or observation

Columns (x_j)

Each column represents a specific attribute or feature

$D = \{e_1, e_2, \dots, e_n\}$
where each $e_i = (v_{i,1}, v_{i,2}, \dots, v_{i,d})$

	x_1	x_2	$\dots$	x_d
e_1	$v_{1,1}$	$v_{1,2}$	$\dots$	$v_{1,d}$
e_2	$v_{2,1}$	$v_{2,2}$	$\dots$	$v_{2,d}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
e_n	$v_{n,1}$	$v_{n,2}$	$\dots$	$v_{n,d}$

Datasets

Examples of datasets

<https://brightdata.com/blog/web-data/what-is-a-dataset>

	A	B	C	D	E	F	G
1	Date	Average price in USD	Total Sold	Smal Avocados Sold	Large Avocados Sold	Extra Large Avocados Sold	City
2	12/27/2022	1.29	319746	292097	27351	298	Atlanta
3	12/20/2022	1.38	272580	251774	20702	103	Atlanta
4	12/13/2022	1.26	356227	324932	31019	276	Atlanta
5	12/6/2022	1.37	306947	283024	23741	182	Atlanta
6	11/29/2022	1.29	279486	250289	28890	308	Atlanta
7	11/22/2022	1.3	300032	269799	29732	501	Atlanta
8	11/15/2022	1.43	292814	263916	28442	456	Atlanta
9	11/8/2022	1.42	285413	250441	34483	488	Atlanta
10	11/1/2022	1.29	353690	290458	62980	253	Atlanta
11	10/25/2022	1.39	301727	236814	64608	304	Atlanta
12	10/18/2022	1.39	288733	228710	59732	291	Atlanta
13	10/11/2022	1.25	347580	255933	91047	600	Atlanta
14	10/4/2022	1.26	359222	265797	92780	644	Atlanta
15	9/27/2022	1.07	324659	262107	61871	681	Atlanta

Datasets

Examples of datasets

https://www.researchgate.net/publication/308698620_FedCSIS_2015_text-preprint

No.	Number of times pregnant	Plasma glucose concentration	Diastolic blood pressure	Triceps skin fold thickness	2-Hour serum insulin	Body mass index	Diabetes pedigree function	Age	Diabetes
1	6	148	72	35	0	33.6	0.627	50	tested_positive
2	1	85	66	29	0	26.6	0.351	31	tested_negative
3	8	183	64	0	0	23.3	0.672	32	tested_positive
4	1	89	66	23	94	28.1	0.167	21	tested_negative
5	0	137	40	35	168	43.1	2.288	33	tested_positive
6	5	116	74	0	0	25.6	0.201	30	tested_negative
7	3	78	50	32	88	31.0	0.248	26	tested_positive
8	10	115	0	0	0	35.3	0.134	29	tested_negative
9	2	197	70	45	543	30.5	0.158	53	tested_positive
10	8	125	96	0	0	0.0	0.232	54	tested_positive
11	4	110	92	0	0	37.6	0.191	30	tested_negative
12	10	168	74	0	0	38.0	0.537	34	tested_positive
13	10	139	80	0	0	27.1	1.441	57	tested_negative
14	1	189	60	23	846	30.1	0.398	59	tested_positive

Datasets

Examples of datasets

https://documentation.sas.com/doc/en/workbenchcdc/v_001/vwblepg/p1r10h01312au5n10blzbelljk7r.htm

Earthquakes by Magnitude							
Obs	Latitude	Longitude	Depth	Magnitude	dNearestStation	RootMeanSquareTime	Type
1	36.8408	-97.875	4.2510	2.5	.	0.4600	earthquake
2	36.0210	-97.097	4.0420	2.5	.	0.6300	earthquake
3	36.7038	-98.041	12.7600	2.5	0.20300	0.2500	earthquake
4	36.7554	-97.556	6.2220	2.5	.	0.7600	earthquake
5	41.9035	-119.662	1.7742	2.5	0.47700	0.2545	earthquake
6	36.3510	-84.995	20.7000	2.5	0.34136	0.2000	earthquake
7	35.7968	-97.444	5.4500	2.5	.	0.3500	earthquake
8	36.7484	-97.649	3.3550	2.5	.	0.5500	earthquake
9	41.8692	-119.615	5.0000	2.5	0.65577	0.6500	earthquake
10	41.8652	-119.673	0.2000	2.5	0.46712	0.4500	earthquake

Datasets

Types of datasets



Structured Data

- Organized in tables/databases with clear schema
- Easy to search and analyze

Examples: relational databases, spreadsheets, CSV files



Unstructured Data

- No predefined format
- Requires significant processing before analysis

Examples: text documents, images, videos, social media posts



Semi-structured Data

- Possesses organizational properties but lacks a rigid schema

Examples: JSON, XML, emails

Datasets

Data quality

Data quality is crucial for reliable analysis and machine learning models, ensuring that insights derived are trustworthy and effective.

Accuracy

Data correctly represents reality

Completeness

All required data is present

Consistency

Data is uniform across datasets

Timeliness

Data is up-to-date and available when needed

Validity

Data conforms to defined formats and rules

Datasets

Dimensionality

What is Dimensionality?

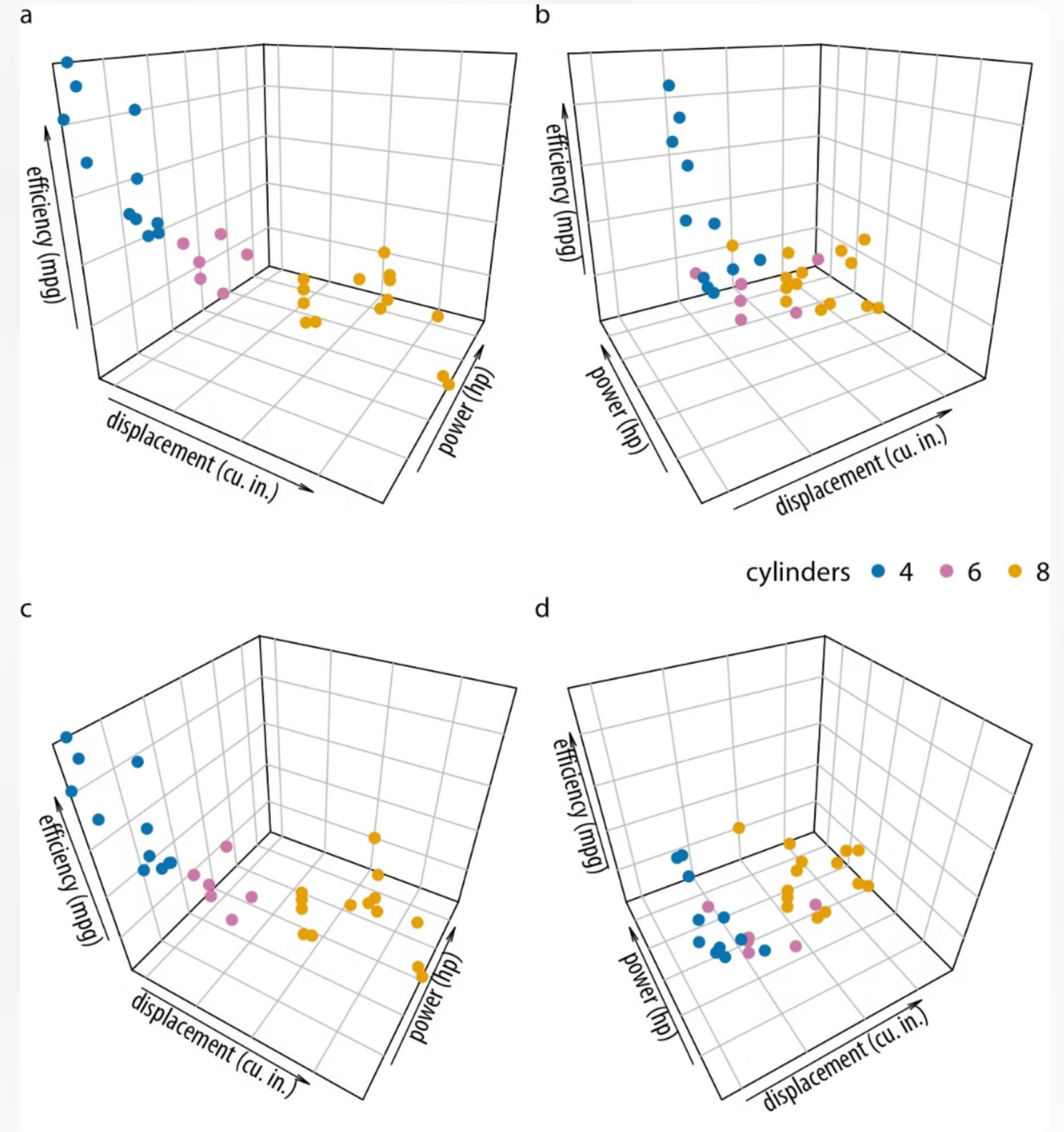
It's the count of variables in a data set

High-Dimensional Data

When the number of variables exceeds the number of examples

Fundamentals of Data Visualization:

<https://clauswilke.com/dataviz/no-3d.html>



Datasets

Dimensionality

When dealing with high-dimensional datasets, creating 2D projections is crucial for effective visualization and analysis

Why 2D Projections?

Simplify complex data, making it easier for humans to visualize and understand

Fundamentals of Data Visualization:

<https://clauswilke.com/dataviz/no-3d.html>

